Latent Volatility Regimes in Cointegrated U.S. Oil Equity Pairs

A Markov Switching Approach

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November 2026

Motivation

- U.S. oil equities share exposure to crude prices, demand and financing conditions, so pairs of them often move together in the long run
- A cointegrated pair has a spread that reverts to an equilibrium
- In stress periods (2008, 2015 to 2016, 2020) spreads become far more volatile
- Two very different things could be happening:
  1. the equilibrium itself has shifted, or
  2. the equilibrium is intact but the shocks around it are larger
- This thesis separates the two by modelling the size of the shocks as a latent regime

Same equilibrium, bigger shocks

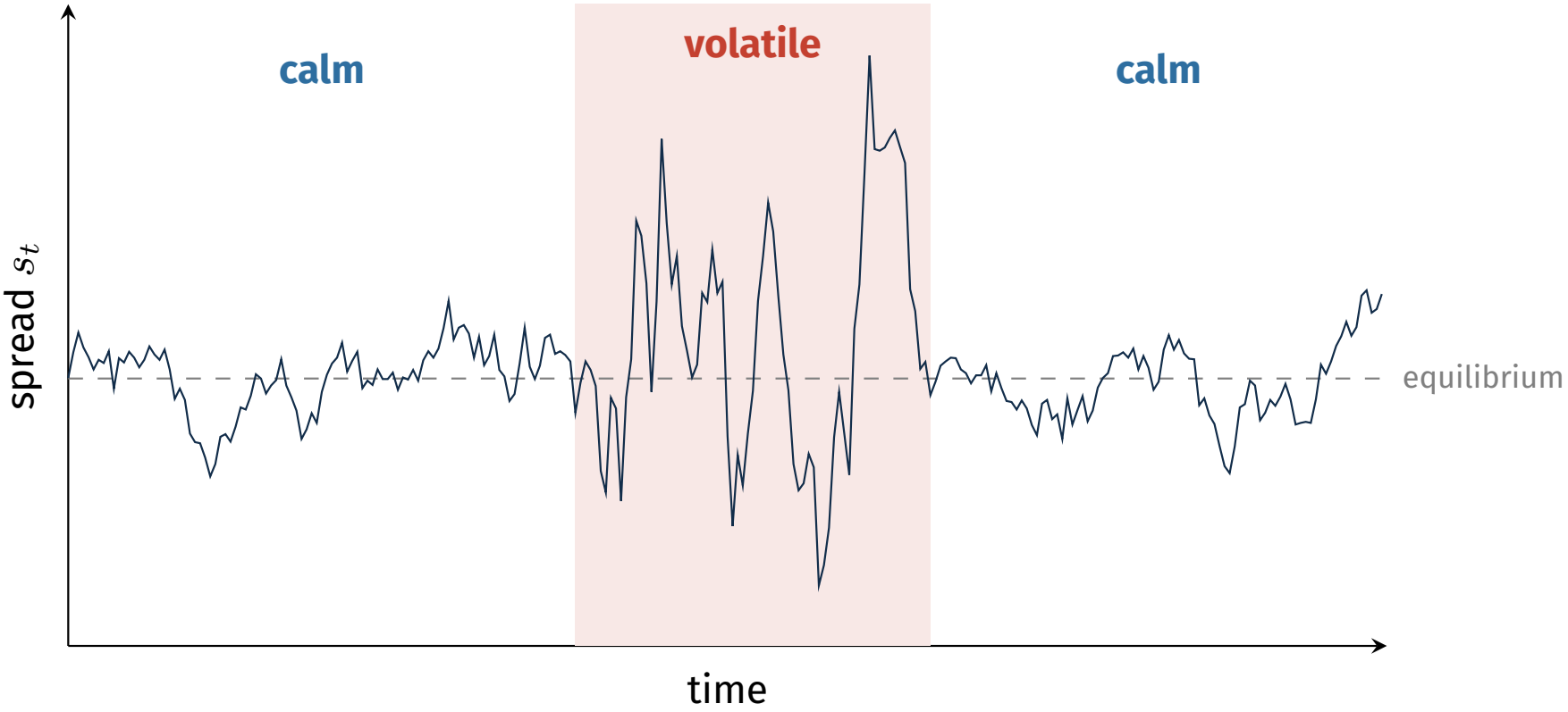


Illustration: the spread keeps reverting to the same level, only the size of the shocks changes.

Research question

Central question

Do the spreads of cointegrated U.S. oil equity pairs switch between calm and volatile regimes, and what do those regimes tell us about how stable the pairs are?

Sub questions

- **Existence.** Is a calm and a volatile state a better description of the spread than a single constant variance?
- **Regime persistence.** Do the estimated calm and volatile states persist through time, and how often does the model switch between them?
- **Synchronisation.** Do many pairs enter the volatile state at the same time?
- **Adjustment.** Does the spread correct more slowly in the volatile state?

Scope: a description of the equilibrium error, not a trading strategy.

Data

- CRSP daily file, January 2000 to December 2025
- Universe: oil producers (SIC 1311, 1321) and refiners and integrated majors (SIC 2911)
- Point in time membership: firms enter at listing and leave at delisting, so no survivorship bias
- Entry filters on the formation window: returns on at least 95% of trading days, price of at least $5 and size above the NYSE 10th percentile
- Prices are total return indices chained from daily returns, so dividends, splits and delisting returns are included
- All models use $\log$ of the total return index

Pair formation

- Rolling design: a three year formation window, stepped forward one month at a time
- Pairs are chosen by economic link, not by correlation
  - producer with producer, refiner with refiner, producer with refiner
  - every such pair is tested, since cointegrated pairs need not have highly correlated returns
- **Engle Granger test** on each pair
  - Estimate long run relationship between each pair using OLS
  - Apply ADF test on the residuals, using MacKinnon p values
  - Run in both directions; the larger p value is kept, so the result does not depend on labelling
  - Classify a pair as cointegrated when the larger p value is $p < 0.05$

Long run relationship

For a pair with log price series y_t and x_t,

$$y_t = \alpha + \beta x_t + s_t$$

- β is the long run elasticity (the hedge ratio)
- s_t is the spread, or equilibrium error
- The pair is cointegrated if $y_t, x_t \sim I(1)$ and $s_t \sim I(0)$
- The spread is computed from the regression in which the larger firm by market capitalisation is x_t

Short run adjustment

With α, β fixed at their formation estimates,

$$\Delta s_t = \mu + \gamma s_{t-1} + \sigma_{S_t} z_t, \qquad z_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$$

- $\gamma < 0$ is the speed of adjustment; half life $= -\ln 2 / \ln(1+\gamma)$
- $S_t \in \{1, 2\}$ is an unobserved state that sets the size of the shock
- Only the shock scale switches. The adjustment speed γ stays fixed within a window

Two state switching variance

The size of the shock depends on a hidden state:

$$\sigma_{S_t} = \begin{cases} \sigma_1 & S_t = 1 \text{ (calm)} \\ \sigma_2 & S_t = 2 \text{ (volatile)} \end{cases} \qquad 0 < \sigma_1 < \sigma_2$$

1. Estimate

On the formation window, the EM algorithm estimates σ_1, σ_2 and how often the states switch

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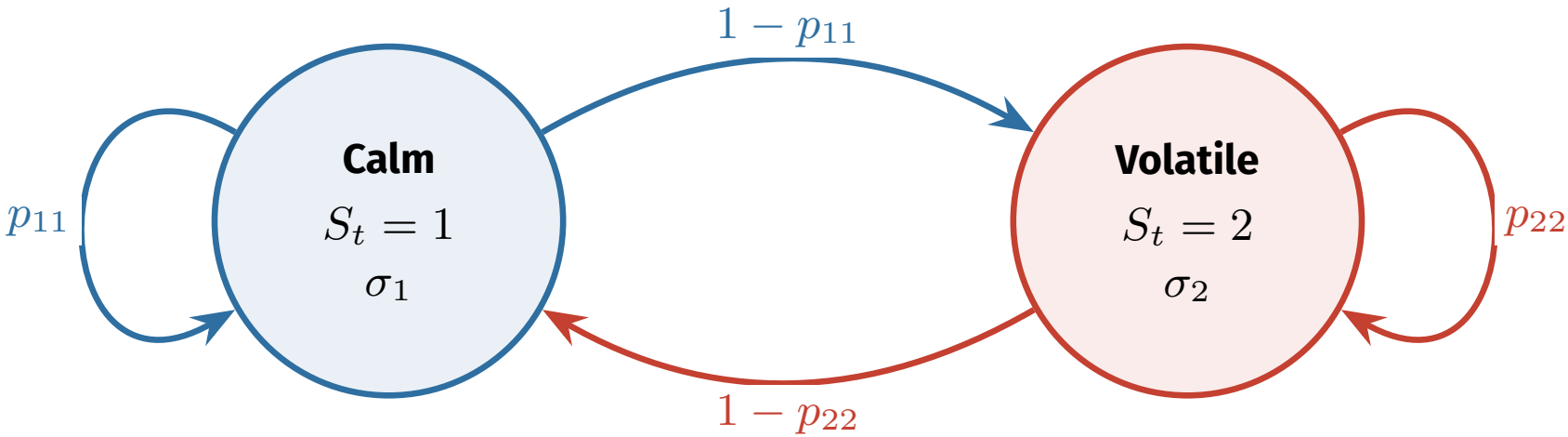
2. Update over time

Each new day, the model updates the chance the spread is volatile, $\Pr(S_t = 2 \mid \mathcal{F}_t)$

The state is never observed directly. The model infers it from how large recent moves in the spread have been.

Key point: a spread can become much more volatile while the equilibrium is unchanged.

Regime dynamics: a two state Markov chain



Transition matrix

$$\mathbf{P} = \begin{pmatrix} p_{11} & 1 - p_{11} \\ 1 - p_{22} & p_{22} \end{pmatrix}$$

Expected duration

$$D_j = \frac{1}{1 - p_{jj}}$$

p_{jj} close to one means long, clustered regimes.

Hypotheses

H1. **Two regimes exist.** The two state model beats a constant variance model. The likelihood ratio has no standard distribution under the null, so it is simulated. A one state Student t is also compared, so fat tails are not mistaken for regimes

H2. **Regimes are persistent.** p_{11}, p_{22} are high, durations are long and σ_2/σ_1 is well above one

H3. **Regimes are sector wide.** The share of pairs in the volatile state rises together in stress episodes rather than pair by pair

H4. **Volatility regimes and equilibrium stability** Pairs characterized by greater high-volatility-state occupancy or persistence are more likely to exhibit subsequent instability in their long-run cointegrating relationship.

Extensions

- **Kalman filter.** Let the long run elasticity drift, $\beta_t = \beta_{t-1} + \eta_t$, and compare regime estimates with a time varying hedge ratio against the static one
- **Conditioning on sector stress.** Transition probabilities that depend on oil price volatility or an oil sector index

Limitations

- **Static long run elasticity.** β is fixed within a window; a genuine shift in the relation shows up as a breakdown rather than as a smooth change
- **Chance passes.** Many tests across overlapping windows mean some pairs pass by luck, and a chance pass can repeat month after month; the chance benchmark measures how often
- **Gaussian states.** Regime separation can be sensitive to outliers and to the normality assumption
- **Corporate events.** Mergers, spin offs and changes in business mix can break a pair for reasons unrelated to volatility

Thank you

Questions